Characterization of Campylobacter spp. transferred from naturally contaminated chicken legs to cooked chicken slices via a cutting board.

Campylobacter represents the leading cause of gastroenteritis in Europe. Campylobacteriosis is mainly due to C. jejuni and C. coli. Poultry meat is the main source of contamination, and cross-contaminations in the consumer's kitchen appear to be the important route for exposure. The aim of this study was to examine the transfer of Campylobacter from naturally contaminated raw poultry products to a cooked chicken product via the cutting board and to determine the characteristics of the involved isolates. This study showed that transfer occurred in nearly 30% of the assays and that both the C. jejuni and C. coli species were able to transfer. Transfer seems to be linked to specific isolates: some were able to transfer during separate trials while others were not. No correlation was found between transfer and adhesion to inert surfaces, but more than 90% of the isolates presented moderate or high adhesion ability. All tested isolates had the ability to adhere and invade Caco-2 cells, but presented high variability between isolates. Our results highlighted the occurrence of Campylobacter cross-contamination via the cutting board in the kitchen. Moreover, they provided new interesting data to be considered in risk assessment studies.